

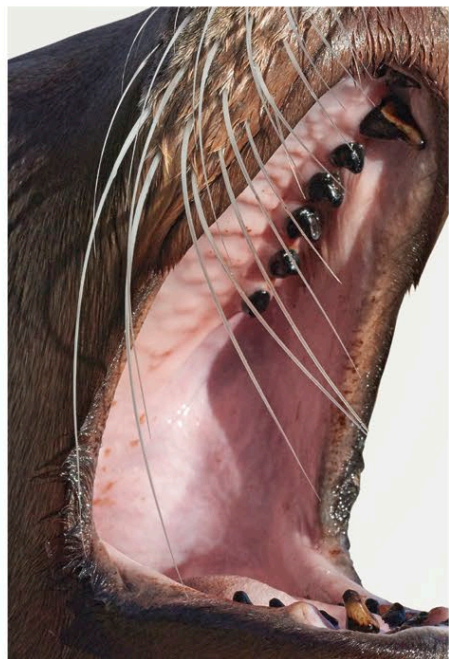
CALIFORNIA SEA LION

Zalophus californianus

The California sea lion lives in polar, temperate, and subtropical waters and on land, and therefore needs to cope with a wide range of environmental conditions. The waters in which it feeds are cold but the beaches on which this sea lion breeds can be hot, so it needs an efficient way of regulating its internal temperature. Its body has an insulating layer of blubber 2.5cm (1in) thick to help keep the sea lion warm when hunting in water and if it becomes too hot on land, it can cool down by returning to the sea. The flippers, which have little fur and lack blubber, have a heat exchange system that can retain or expel body heat by transferring it between arteries and veins at the flipper bases and by controlling blood flow to the skin. The fore flippers may also be waved in the air to help with cooling – a behaviour seen both in water and on land. California sea lions are hunted by sharks and orcas.

EAR >

Unlike seals, sea lions have small, pointed, external ear flaps that fold down and back against the head. They can hear well both on land and under water.

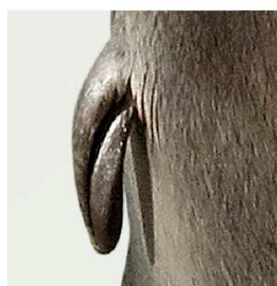
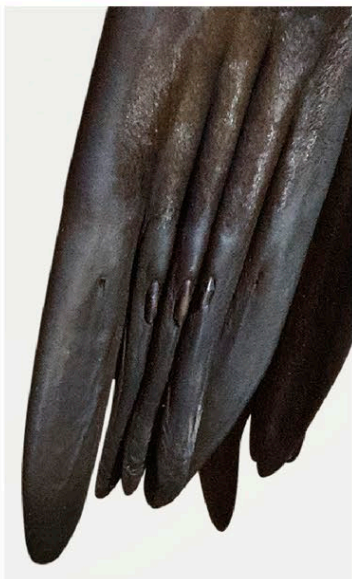


^ MOUTH

The mouth contains 34 teeth. Initially white, they turn black with increasing age. The four large, recurved canines are used to grasp and hold on to prey.

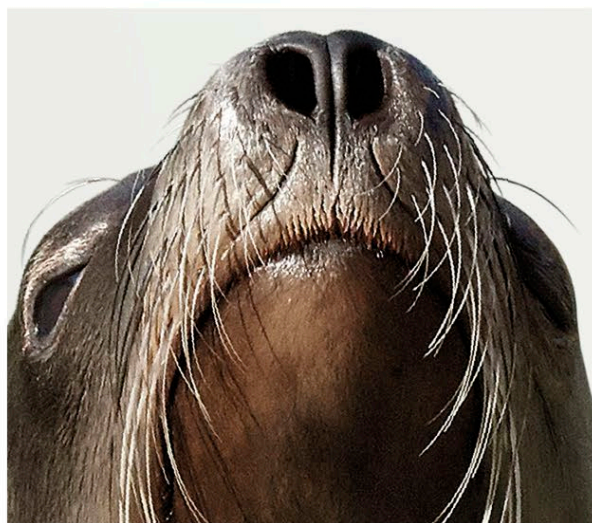
HIND FLIPPER >

In water, the hind flippers trail behind the body and are used for steering, but for moving on land they are rotated forwards and used with the front flippers.



^ DIVING

Dives last for up to ten minutes during which the sea lion's heart rate falls to about 20 beats per minute.



^ NOSE

A sea lion's nostrils are only opened when inhaling or exhaling. The sensory whiskers help in working out the size and shape of objects when under water.

^ FORE FLIPPER

Swimming is powered by the long fore flippers, which enable the sea lion to achieve speeds of over 21 kph (13 mph).



flexible back gives the sea lion agility and manoeuvrability when hunting fish

streamlined body reduces drag

gently sloping forehead

black, leathery nose pad

> FEMALE SEA LION

California sea lions are sexually dimorphic. Females are smaller and less muscular, particularly around the neck and shoulders, and their fur is lighter. They also have a gently sloping forehead compared to the more upright one of males.

